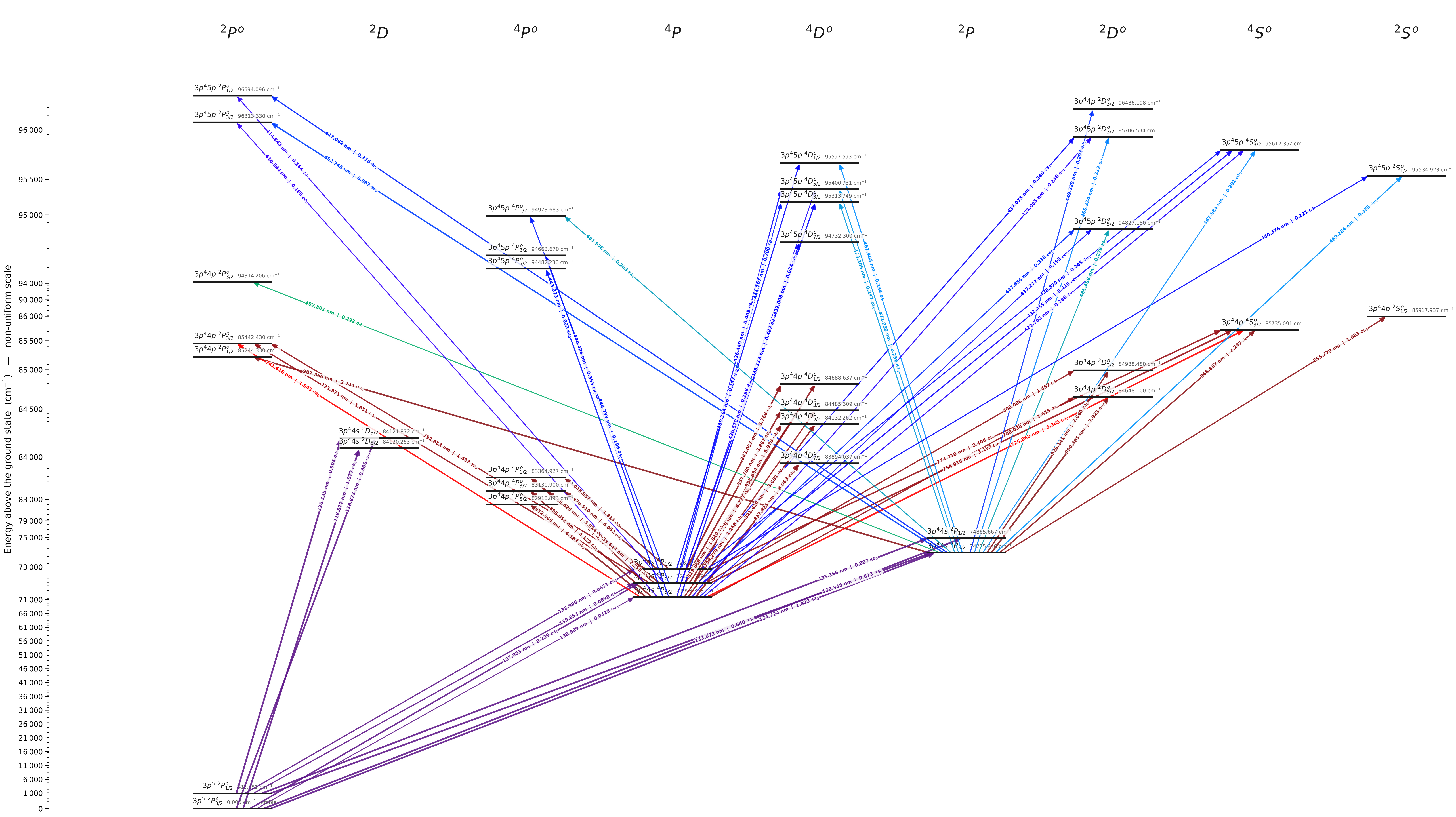
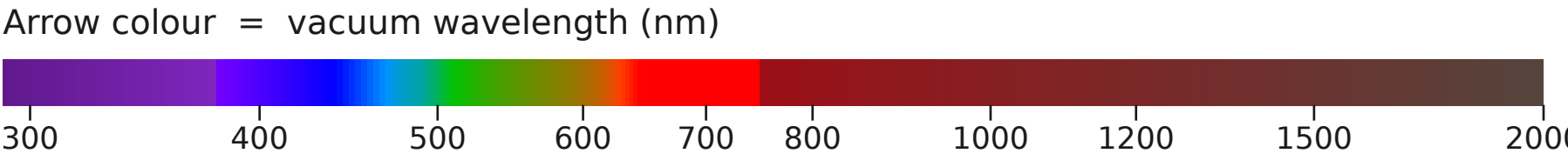


Chlorine

Cl (Cl I). Strongest classified transitions below 2 μm (NIST): 37 levels, 70 lines.

Cl⁺ ionisation limit 104591.01 cm⁻¹ = 12.96763 eV (95.611 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[U||e||J']$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$[U e J']$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3p ⁵ 2p _{3/2} ^o - 3p ⁴ 4s 2D _{3/2}	118.8751	-	2521.9103	0.300	2.710e+07	-	-	
3p ⁵ 2p _{3/2} ^o - 3p ⁴ 4s 2D _{5/2}	118.8774	-	2521.8620	1.077	2.330e+08	-	-	
3p ⁵ 2p _{1/2} ^o - 3p ⁴ 4s 2D _{3/2}	120.1352	-	2495.4581	0.904	2.390e+08	-	-	
3p ⁵ 2p _{3/2} ^o - 3p ⁴ 4s 2P _{1/2}	133.5726	-	2244.4162	0.640	1.740e+08	-	-	
3p ⁵ 2p _{3/2} ^o - 3p ⁴ 4s 2P _{3/2}	134.7240	-	2225.2349	1.422	4.190e+08	-	-	
3p ⁵ 2p _{1/2} ^o - 3p ⁴ 4s 2P _{1/2}	135.1656	-	2217.9640	0.887	3.230e+08	-	-	
3p ⁵ 2p _{1/2} ^o - 3p ⁴ 4s 2P _{3/2}	136.3447	-	2198.7827	0.613	7.500e+07	-	-	
3p ⁴ 4s 4P _{3/2} - 3p ⁴ 4p 4S _{3/2}	725.8618	725.6618	413.0159	3.365	1.500e+07	-	-	
3p ⁴ 4s 4P _{3/2} - 3p ⁴ 4p 4D _{3/2}	833.5597	833.3307	359.6533	4.277	1.600e+07	-	-	
3p ⁴ 4s 4P _{3/2} - 3p ⁴ 4p 4D _{5/2}	837.8245	837.5943	357.8225	8.063	2.800e+07	-	-	
3p ⁴ 4s 4P _{1/2} - 3p ⁴ 4p 4D _{3/2}	843.0567	842.8250	355.6018	3.768	2.400e+07	-	-	
3p ⁴ 4s 4P _{3/2} - 3p ⁴ 4p 4D _{5/2}	858.8340	858.5981	349.0692	5.970	1.900e+07	-	-	
3p ⁴ 4s 2P _{3/2} - 3p ⁴ 4p 2P _{1/2}	907.5659	907.3168	330.3258	3.744	1.900e+07	-	-	
3p ⁴ 4s 4P _{3/2} - 3p ⁴ 4p 4P _{3/2}	912.3646	912.1143	328.5884	6.183	1.700e+07	-	-	
3p ⁴ 4s 4P _{3/2} - 3p ⁴ 4p 4P _{1/2}	919.4253	919.1731	326.0650	4.014	2.100e+07	-	-	
3p ⁴ 4s 2P _{3/2} - 3p ⁴ 4p 2D _{5/2}	959.4853	959.2222	312.4513	7.923	2.400e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).